MILITARY SPECIFICATION

COMPOUND, ASPHALTIC, HIGH MELTING HOT-MELT (CAVITY LINING)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers a high-melting hot-melt asphaltic compound for coating the inner walls of metal cases prior to filling with explosives.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

SS-R-406	Road and Paving Materials: Methods of Sampling and Testing
PPP-D-723	Drums, Fiber

Military

MIL-T-248	Trinitrotoluene (TNT)
MIL-C-450	Coating Compound, Bituminous Solvent Type
MIL-E-22267	Explosive Compositions, HBX Type
MIL-A-512	Aluminum, Powdered, Flake, Grained, and and Atomized

FSC 8030

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STANDARDS

Federal

FED-STD-313	Symbols For Packages and Containers For Hazardous Industrial Chemicals and Materials

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-290	Packaging, Packing and Marking of Petroleum and Related Products

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

D 36	Softening Point of Asphalts and Tar Pitches (Ring-and-Ball Apparatus)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.

3.1.1 Qualification. The asphaltic compound furnished under this specification shall be a product which meets all the requirements specified herein and has been listed on or approved for listing on the applicable qualified products list at the time set for opening of bids. A change in formulation in either materials or amounts of materials shall require requalification of the product.

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3.1.2 Materials. The compound shall be an asphaltic compound. Use of a polyamide wax as an additive to adjust viscosity of the molten compound shall be permissible. The materials used in the manufacture of the asphaltic compound shall be of acceptable quality suitable for the purpose intended and shall be such as to produce a compound which is easily heat-liquifiable but does not flow at temperatures slightly below the melting point. The compound shall contain no volatile solvents.

3.1.3 Conflicting requirements. Conflicting requirements arising between this specification and any specification, publication or inspection requirement listed herein shall be referred in writing to the cognizant agency (6.3.1), or appointed agent, for interpretation, clarification, and resolution.

3.1.4 Workmanship. The asphaltic compound ingredients shall be processed in a manner that will produce the highest quality material necessary to meet the requirements of this specification. The asphaltic compound shall be homogeneous and free from grit, foreign matter, lumps, or other defects that could adversely affect its intended use. The asphaltic compound shall be capable of storage at ambient temperature without release or separation of any component.

3.2 Product characteristics.

3.2.1 Chemical requirements. The asphaltic compound shall conform to the chemical requirements specified in table I when tested in accordance with 4.6.1.1 through 4.6.1.5.

Table I

CHEMICAL REQUIREMENTS

	Percent Maximum
*Manganese	0.10
*Lead	0.04
Acidity, as H_2SO_4	0.01
Alkalinity, as NaOH	0.01
Insoluble matter in CCl_4	1.00
*Ash	1.00

*Note: Quality conformance tests (4.6.1.1 and 4.6.1.2) for manganese and lead are not required when the ash content is less than 0.004 percent.

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3.2.2 Viscosity. The viscosity of the asphaltic compound shall be a maximum of 25 Zahn seconds at 232° ± 2° Celsius (C) when tested in accordance with 4.6.3.

3.2.3 Softening point stability. The softening point of the asphaltic compound shall be within the temperature range of 172° to 205° C when tested in accordance with 4.6.4.

3.2.4 Penetration. The asphaltic compound shall show the following penetration values when tested in accordance with 4.6.5:

(a)	200 g/60 sec/0° C	5 - 35
(b)	100 g/5 sec/25° C	5 - 35
(c)	50 g/5 sec/46° C	80 max.

3.2.5 Flash point. The minimum flash point of the asphaltic compound shall be 300° C when tested in accordance with 4.6.6.

3.3 Performance requirements.

3.3.1 Heat resistance. A dried film of the asphaltic compound on a metal panel shall not sag or flow when heated by immersion in a silicone oil bath maintained at 172° ± 1° C for 1 hour. Test procedure shall be in accordance with 4.6.7.

3.3.2 Acid resistance. A dried film of the asphaltic compound on a metal panel shall withstand the action of picric, sulfuric, nitric, and hydrochloric acids for 6 hours without disintegration, browning, or dulling. There shall be no etching of the metal underneath the compound. Test procedure shall be in accordance with 4.6.8.

3.3.3 Water resistance. A dried film of the asphaltic compound on a metal panel shall withstand immersion in distilled water for 18 hours without blistering, whitening, and not more than a slight dulling. Test procedure shall be in accordance with 4.6.9.

3.3.4 Flexibility. A dried film of the asphaltic compound on a metal panel shall show no chipping, scaling, or cracking when tested in accordance with 4.6.10.

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3.4 Reactivity requirement.

3.4.1 Reactivity. The reactivity for the asphaltic compound in contact with H-6 (MIL-E-22267) explosive and tritonal (80 ± 3 parts TNT, MIL-T-248, and 20 ± 3 parts aluminum, MIL-A-512) explosive in the vacuum stability test of 4.6.11 shall not exceed 3.0 milliliters of gas (over and above that generated by the controls).

3.5 Application - coating requirements.

3.5.1 The asphaltic compound shall be suitable for application into bombs using equipment and processing techniques that provide the interior coating requirements of applicable drawings for thermally protected bombs (see 6.3.2) in one application. The applied coating shall be self adhering and also shall adhere to surfaces coated with cavity paint, MIL-C-450. Any inherent separation of components of the asphaltic compound shall not affect application or coating characteristics of the material.

3.5.2 The coated bombs (3.5.1) shall be capable of being loaded with H-6 explosive to the requirements of applicable drawings for thermally protected bombs (see 6.3.2) without separation of the coating or other loss of coating effectiveness.

3.5.3 Bombs coated with the asphaltic compound (3.5.1) and loaded with H-6 explosive (3.5.2) shall survive the fast cookoff test of 4.6.12. The cookoff time shall be greater than 4.0 minutes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all quality conformance inspection requirements specified herein except the reactivity requirement of 3.4. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the asphaltic compound shall be classified as follows:

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(a) Qualification Inspection (4.3)
(b) Quality Conformance Inspection (4.4).

4.3 Qualification inspection. The qualification inspection shall consist of all the tests of this specification as described under test methods and procedures (4.6). The qualification sample (4.3.1) shall meet all the requirements of this specification. The cognizant agency (6.2.1) reserves the right to accept the manufacturer's certification of testing, when available, in lieu of Government testing.

4.3.1 Qualification sample. The qualification sample shall consist of 3000 pounds of the asphaltic compound manufactured and taken from one batch of the material. The material shall be packaged in suitable containers and forwarded to the Commanding Officer, Naval Weapons Station, Yorktown, Va. 23491, Attention: NEDE Dept. Code 502. Each sample submitted shall be plainly identified by securely attached, durable labels marked with the following information:

(a) Compound, asphaltic, high melting, hot-melt (cavity lining)

(b) Formulation and identity of additives, if any, to the asphaltic material

(c) Name and address of manufacturer

(d) Location and identity of the plant which produced the samples

(e) Manufacturer's identification

(f) Date of manufacture

(g) Submitted by (name) (date), for qualification tests in accordance with the requirements of MIL-C-82604.

4.3.2 Fast cookoff samples. Three bombs, coated and explosive loaded to the requirements of 3.5.1 and 3.5.2 respectively, shall be forwarded to an activity designated by the cognizant agency for the fast cookoff test of 4.6.12. The cognizant agency shall designate the loading activity for preparing these samples.

4.4 Quality conformance inspection. For each inspection lot of material submitted for acceptance, quality conformance inspection shall consist of an examination for acceptability of quality control methods used by the manufacturer, an examination of filled containers

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(4.4.3) for conformance to packaging, packing, and marking requirements, and examinations and tests of samples (4.4.2) for all the requirements of 3.1.2 through 3.3.4 of this specification. The reactivity requirement of 3.4 may be a requirement for quality conformance inspection if at any time during a contract the cognizant agency determines the material submitted under this specification to be suspect.

4.4.1 Inspection lot. An inspection lot shall consist of the same type asphaltic compound produced by one manufacturer, at one plant, from the same materials and formulation, and under essentially the same manufacturing conditions.

4.4.2 Sample for tests. A 10-pound sample of asphaltic compound shall be taken at random from each lot in accordance with method 101.11 of SS-R-406. The sample material shall be chosen at random from containers of "solidified" compound. A sample for the tests of 4.6.3 through 4.6.10, herein, shall be preconditioned for testing in accordance with 4.6.2. The sample for tests of 4.6.1 need not be preconditioned. The lot shall be unacceptable if it fails to meet all the requirements of this specification, except as noted in 4.4.

4.4.2.1 Sampling. Each container shall be labeled with complete information on the lot and batch number, date of sampling, contract number, and applicable specification.

4.4.3 Sample for examination of filled containers. A random sample of filled containers shall be selected from each lot of compound in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) - 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 General. Unless otherwise specified, the values specified in section 3 apply to the results obtained from one determination for each particular test.

4.5.2 Visual examination. Conformance of the asphaltic compound to the requirement for workmanship (3.1.4) and any other requirements not covered by specific tests shall be determined by appropriate visual examination.

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4.5.3 For examination of preparation for delivery. Filled and closed containers and the preparation for delivery thereof shall be examined for compliance with all requirements of this specification in regard to contents, closure, damaged or leaking container, improper container, and marking. The inspection shall be level I and the AQL shall be 2.5 percent defective. The sample containers shall be shipped as part of the lot if the lot is accepted.

4.6 Test methods and procedures.

4.6.1 Chemical requirements.

4.6.1.1 Manganese. Ash a 10-gram sample of the compound, dissolve in 10 milliliters (ml) of concentrated nitric acid, and filter if necessary. Dilute to 100 ml with distilled water and transfer a 20-ml aliquot of the solution to a 250-ml beaker. Add 1 gram of lead peroxide and boil. Note if manganese is present as indicated by the development of a red coloration in the solution. If the presence of manganese is indicated, filter through asbestos and titrate the clean filtrate with N/10 oxalic acid until the red coloration disappears. Calculate the percent manganese in the sample as follows:

$$\text{percent manganese} = \frac{1.099VN}{W}$$

where:

V = milliliters of N/10 oxalic acid used
N = normality of oxalic acid
W = weight of sample.

4.6.1.2 Lead. Transfer the remaining 80-ml portion of the solution obtained as described in paragraph 4.6.1.1 to a beaker and evaporate to 5 ml. Make the solution slightly alkaline with ammonium hydroxide. Acidify with acetic acid and add an excess. Bring to the boiling point and add 10 to 15 ml of a 10 percent solution of sodium or potassium dichromate. Boil the solution and allow to stand for 16 hours. Note if lead is present as indicated by a yellow precipitate. If lead is present, filter the solution through a tared Gooch crucible (or equivalent) and wash the precipitate with water and alcohol. Dry the crucible and contents at approximately 110° C for 1 hour, cool, and weigh. Calculate the increase in weight of the crucible as lead in the sample as follows:

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$$\text{percent lead} = \frac{64.1A}{W}$$

where:

A = weight of residue
W = weight of sample.

4.6.1.3 Acidity or alkalinity. Dissolve a 5-gram sample of the compound in 10 ml of toluene which is neutral to phenolphthalein. Shake the solution with 100 ml of distilled water for at least 5 minutes. Filter and titrate with N/100 H_2SO_4 or N/100 NaOH as necessary, using phenolphthalein as indicator. A blank shall be run on an equal volume of the water used.

4.6.1.4 Insoluble matter in CCl_4.

4.6.1.4.1 Apparatus. The extraction apparatus shall consist of a heat resistant glass assay flask, a block tin condensing coil with a nickel, copper or tin-plated metal cap which fits securely around the rim of the flask, and a glass siphon cup which is suspended under the condenser. This apparatus shall be suitable for use with an electric hot plate or steam bath.

4.6.1.4.2 Method. Transfer a sample of approximately 6 grams, weighed accurately to 0.1 mg into a tared stainless steel 200-mesh thimble. Fabricate thimble from a 15-centimeter (cm) diameter circle of wire mesh by using the lower portion of a test tube as a form for folding the thimble. This form should be of a diameter that just will fit into the glass siphon tube. Place thimble containing the sample into the siphon cup. The thimble should not extend beyond the top of the cup. Suspend the cup under the condenser by a fine wire bail. Add approximately 60 ml of carbon tetrachloride to the sample in the cups; allow the excess solvent to siphon over into the flask. Put condenser in place and circulate cold water through it. Heat the flask on a steam bath or an electric hot plate. Solvent should drip from the top of the condenser at a slow to medium rate. The extraction shall be discontinued when the solvent drops colorless from the siphon cup. Wash down the steel mesh with a fine jet of solvent from a wash bottle. If the washings show any color, the sample shall be put back into the siphon cup and the extraction continued until the solvent again becomes colorless. Remove the steel mesh thimble containing the residue, dry for 1 hour at approximately 100° C, cool, and weigh. Calculate the percent insoluble in the solvent as follows:

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$$\text{Percent insoluble in carbon tetrachloride} = \frac{A - B}{W} \times 100$$

where:

A = weight of steel mesh thimble + residue after extraction in grams
B = weight of steel thimble in grams
C = weight of sample in grams.

4.6.1.5 Ash. Transfer a weighed portion of approximately 1 gram of the sample to a tared procelain or platinum crucible and place in a cold muffle furnace having good air circulation and capable of temperature regulation between 700° and 750° C. Gradually raise the temperature of the furnace to a red heat. Finish the ignition to constant weight at a temperature between 700° and 750° C. Cool in a desiccator and weigh. Calculate the weight of residue as percent ash in the sample.

4.6.2 Test sample preparation. The material for all of the tests required by paragraphs 3.2.2 through 3.3.4 shall be prepared at the same time and from the same melt of the test sample selected in accordance with 4.4.2 as follows.

4.6.2.1 Apparatus.

(a) Sample container - Stainless steel, 1-liter beaker approximately 4-1/8 inches I.D. by 5-3/4 inches high, straight sides, no pouring lip.

(b) Oven - Electric, mechanical convection type, temperature range: ambient to 250° C.

(c) Stirring rod - 5/16-inch diameter, 12 inches long, Teflon with a steel core.

4.6.2.2 Preconditioning. A 600- to 650-gram sample of the asphaltic compound selected in 4.4.2 shall be transferred to a 1-liter stainless steel beaker. The sample shall be heated in an oven, mechanical convection type, for 3-3/4 hours. The temperature of the oven shall be 232° ± 2° C. The oven shall be at temperature when the sample is placed in it. The sample shall be located in the center of the oven with good air circulation on all sides. Sample shall be hand stirred with the Teflon stirring rod every 30 minutes to prevent lumping. At the end of the heating period the sample shall be removed from the oven for the Zahn viscosity test of 4.6.3. When this test is completed, the

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sample and its container shall be returned to the same oven and heated for an additional 30 minutes. Sample shall remain out of the oven for viscosity testing no longer than 2 minutes. At the end of the second heating period, samples shall be poured and steel panels coated for the tests of 4.6.4, 4.6.5, 4.6.6, 4.6.7, 4.6.8, 4.6.9, and 4.6.10.

4.6.3 Viscosity. Determine the viscosity of the asphaltic compound with a Zahn viscosimeter, No. 5 cup, at the end of the 3-3/4 hours of heating at 232° ± 2° C. Preheat the viscosimeter in the same oven and at the same time as the sample is being prepared. Remove the sample beaker containing the molten compound and the viscosimeter from the oven. Hold the viscosimeter in a vertical position and immediately dip it into the sample completely filling the cup. Lift the cup one-half inch above the top surface of the beaker and allow the asphaltic compound to drain into the beaker. With a stop watch, record the time in seconds between when the cup breaks the surface of the sample and when the steady flow of the liquid from the orifice suddenly stops.

4.6.4 Softening point stability. Softening point shall be determined on two samples run in the same bath at the same time in accordance with method 216.0 of SS-R-406 using the double ring holding device of ASTM D 36 except Dow Corning Silicone Fluid DC-200 of 350 centistokes viscosity shall be used in lieu of water for the bath fluid; thermometer shall be a total immersion, range -10° to 250° C, 0.5° divisions, 16 inches in length (Arthur H. Thomas No. 9296-L40 is satisfactory); two shouldered rings shall be employed, and the samples shall be heated at a rate of 5° C/min ± 1° C for the last 30° before the softening point is reached. (Temperature program controller, Model No. 240, Mfg. Hewlett Packard; Chromalox, 2000-watt hot plate, Mfg. Edwin L. Wiegard Co., can be used to obtain the proper heating rate.) The top cover of the ring holding device shall be drilled to accommodate a small glass, motor-driven stirrer with a 6-inch shaft. The shouldered rings shall be heated to 232° ± 2° C before the molten compound is poured. To facilitate release of filled rings from the bottom plate of the mold, wipe the plate with silicone oil, leaving a thin film before pouring the sample. After samples have solidified, place mold on a hot plate. When mold becomes hot, but still can be handled with the bare hand, remove from the hot plate and slide rings from the bare plate to a piece of aluminum foil. Softening point shall be taken as the average of the two values obtained.

4.6.5 Penetration. The penetration values shall be determined in accordance with method 214.01 of SS-R-406. Ointment cups shall be preheated before pouring samples into them.

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4.6.6 Flash point. The flash point shall be determined in accordance with method 217.01 of SS-R-406.

4.6.7 Heat resistance.

4.6.7.1 Preparation of test panel. A solvent-cleaned, mechanically-polished, plain carbon steel panel, 6 × 4 inches, by approximately 1/8-inch thick shall be used. Preheat the panel for 5 minutes, prior to dipping in the preconditioning oven. (See 4.6.2.2.) Coat the panel by dipping it into the molten compound which shall be maintained at 232° ± 2° C. Leave the top portion of the panel uncoated. The asphalt thickness shall be approximately 1/16 inch on each side of the panel. Suspend the panel with the uncoated portion up and allow to cool to room temperature. Trim the excess compound from the bottom edge of the panel with a sharp knife.

4.6.7.2 Method. Suspend a panel, prepared as described in 4.6.7.1, in a bath of silicone oil, Dow Corning, DC-200 (350 cs), at 172° ± 1° C for 1 hour so that the uncoated end of the panel is uppermost and one half of coated portion is submerged. Remove panel, and cool by immersion in cold water. Examine the film for sag or flow. A slight surface displacement at the silicone line is permissible.

4.6.8 Acid resistance. Place a panel, prepared as described in 4.6.7.1, on a flat horizontal surface. Place on the coating, at different places, drops of the following solutions:

Saturated picric acid
Sulfuric acid, sp. gr. 1.30
Nitric acid, sp. gr. 1.22
Hydrochloric acid, sp. gr. 1.09

Allow the acids to remain in contact with the coating for 6 hours while the temperature is maintained at approximately 21° C. Cover the drops with small watch glasses in order to prevent evaporation. Remove the acids by washing with cold water and dry the film at room temperature for 1 hour. Examine the areas previously in contact with the acids and note if the compound is resistant to each of these acids as indicated by absence of disintegration, browning, or dulling of the coating. Do not consider a slight bloom around the area of a spot exposed to acid as an indication of failure to resist the acid. Remove the coating with benzene and examine the metal. Note if the coating has resisted the action of each acid as indicated by the absence of any corrosion of the metal.

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4.6.9 Water resistance. Suspend a panel, prepared as described in 4.6.7.1, in a beaker of cold distilled water for 18 hours so that the uncoated end of the panel is uppermost and one half of the coated portion is submerged. Examine the panel for blistering, whitening, softening, or other visible defects.

4.6.10 Flexibility. A strip of No. 30 gage hot rolled steel sheet, 2 by 8 inches, shall be cleaned by washing with benzol and thoroughly dried. A preheated strip shall be coated on one side to a thickness of 1/16 inch by flowing on the molten compound and then cooled at room temperature for a period of 6 hours. With the coating on the outside, bend the test panel rapidly 180° over a mandrel 1 inch in diameter. No macroscopic cracking of any kind shall be visible after the bending test.

4.6.11 Reactivity. (Qualification test by designated activity only unless otherwise specified.) Determine the reactivity of the asphaltic compound in contact with H-6 and tritonal explosives in accordance with the vacuum stability test procedures of 4.6.11.1 through 4.6.11.4.

4.6.11.1 Preparation of samples. Heat the compound until pourable, then deposit on glass plates a sufficient amount of the compound to provide 2.5 (N+2) grams of dried film (where N equals the number of explosives involved). Air dry the films under ambient conditions for 48 hours. The films are then removed and cut into approximately 1/4-inch squares. It is important that the room in which this operation is conducted is free from acid, alkaline, or other contaminating fumes. Reduce the explosive materials to 12 mesh or finer by grinding or rasping. Keep both the compound film and the ground explosives dry by storing in a desiccator over a desiccating agent until ready for testing.

4.6.11.2 Calibration of apparatus. Calibrate the necessary number of vacuum stability test assemblies, as shown in figure 1, method 403.1.2 of MIL-STD-286.

4.6.11.3 Testing procedure. Use 2 N + 2 (where N equals the number of explosives used) tubes similar to the heating tube portion of the apparatus shown in figure 1, method 403.1.2 of MIL-STD-286. For controls add 2.5 grams of the dried compound to each of two tubes and 2.5 grams of each explosive to additional individual tubes. Place uniform mixtures of 2.5/2.5 grams of the asphaltic compound and each of the explosives specified in the test in single separate tubes. Uniform distribution or mixing is

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obtained by carefully layering alternate portions of the dried compound with the explosive. After the addition of each portion, lightly tamp the mixture with a glass or metal rod to insure intimate contact of the materials. After all the samples have been prepared, connect the respective heating tubes with the capillary tubes as prescribed and conduct the vacuum stability test according to method 403.1.2 of MIL-STD-286 at 100° ± 0.5° C for 48 hours. Make all readings at room temperature with the samples removed from the bath. Correct the readings of both the controls and the test samples to standard conditions of temperature and pressure, and check for compliance with 3.4.

4.6.11.4 Calculation of reactivity. Calculate the reactivity of each of the explosive materials with the coating compound as follows:

$$\text{Reactivity in milliliters of gas} = X - (Y + Z)$$

where

X = ml of gas produced by the mixture of explosive material and asphaltic compound

Y = ml of gas produced by the explosive material alone

Z = ml of gas produced by the asphaltic compound alone.

4.6.12 Fast cookoff test. The three sample bombs (4.3.2) meeting the application-coating requirements of 3.5 shall survive the fast cookoff test defined by the Naval Weapons Laboratory, Dahlgren, Va.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. The material shall be packed in type I, grade C fiber drums conforming to PPP-D-723. The inside of the container shall be treated with a suitable release agent which will not contaminate the contents and which will promote ready removal of the contents. Each drum shall be furnished with a top heading. Drums shall be palletized in conformance with MIL-STD-147.

5.2 Marking. Unless otherwise specified in the contract or order, marking shall be in accordance with MIL-STD-290, as applicable.

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5.2.1 Hazardous material identification. In addition to the marking required by MIL-STD-290, the fiber drums shall be marked with hazardous material identification symbols in accordance with FED-STD-313.

5.2.2 Safety data sheet. When required by the contract or purchase order, the supplier shall prepare a safety data sheet in accordance with FED-STD-313. The safety data sheet shall be submitted with each shipment of asphaltic compound as specified in FED-STD-313.

6. NOTES

6.1 Intended use. The high melting, hot-melt asphaltic compound covered by this specification is intended for use as a protective lining for fabricated metal cases in applications where charges of high explosives are poured into the cases and require thermal insulation from the elevated temperature environment of a fuel fire.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, revision letter, and date of this specification

(b) Quantity of asphaltic compound ordered

(c) Testing activity designated to perform qualification tests

(d) Type and degree of contractor quality assurance program required.

6.2.1 Certification. In lieu of the testing required by 4.3, the supplier may certify that material supplied meets all requirements set forth herein. The supplier shall, at the time of delivery and if requested by the procuring activity, furnish certified copies of test data to the procuring activity. The certification shall include data which compare the performance of the asphaltic compound with all the requirements of this specification.

6.2.2 Contract data requirements. Data specified in paragraphs 5.2.2 and 6.2.1 and test data required to meet the quality conformance testing requirements of paragraph 4.4 shall be delivered as identified on a numbered DD Form 1664 when specified in a DD Form 1423 incorporated into the contract.

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PREVIOUS EDITION IS OBSOLETE

6.3 Definitions.

6.3.1 Cognizant agency. The cognizant agency as defined in this specification is the NEDE Department, Code 502, Naval Weapons Station, Yorktown, Va. 23491.

6.3.2 Thermally protected bombs. Thermally protected bombs as defined in this specification are bombs coated, prior to explosive loading, with insulating materials--exterior or interior or combination of both--that provide a heat resistant barrier between the explosive charge and the elevated temperature environment of a fuel fire. The coating thickness utilized on bombs explosive loaded to meet the bomb application--coating requirements (3.5) of this specification shall be determined by the cognizant agency.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ordnance Systems Command, Department of the Navy, Washington, D.C. 20360; however, information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Weapons Station, Yorktown, Va. 23491, Attention: NEDE Department (Code 502).

6.5 Supersession data. This specification includes the requirements of Purchase Description WS 12675 dated 24 April 1969.

Preparing activity:
NAVY (OS)
(Project No. 8030-N049)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-82604 (OS), Compound, Asphaltic, High Melting Hot-Melt (Cavity Lining)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Attn: Standardization Division (QA3)
Indian Head, Maryland 20640

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